

MAT 201 – Linear Algebra

Spring 2026

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Course Basics				
Credit Hours	3			
Lectures	Number of Lectures in a Week	1	Duration	180 min
Tutorials	Number of Tutorials in a Week (after first 4 lectures)	0	Duration	

COURSE DESCRIPTION

Algebra of linear transformations and matrices. determinants, rank, systems of equations, vector spaces, orthogonal transformations, linear dependence, linear Independence and bases, eigenvalues and eigenvectors ,characteristic equations, Inner product space and quadratic forms.

COURSE PREREQUISITE(S)

- Calculus & Analytical Geometry.
Students should have prior knowledge of basic algebra and analytical mathematics.

COURSE OBJECTIVES

This course aims to develop a comprehensive understanding of linear algebra and its theoretical foundations. Students will acquire the knowledge and analytical skills necessary to work with matrices, linear transformations, determinants, vector spaces, eigenvalues, eigenvectors, and inner product spaces. The course emphasizes both conceptual understanding and problem-solving abilities, enabling students to apply linear algebraic techniques in mathematics, data science, engineering, computer science, and related disciplines.

Learning Outcomes

Upon successful completion of this course, students will be able to demonstrate understanding of matrices and linear transformations, including matrix operations and their algebraic properties. Compute and interpret determinants, rank, and solutions of systems of linear equations using appropriate analytical techniques. Analyze vector spaces, subspaces, linear dependence, linear independence, and bases. Determine eigenvalues, eigenvectors, and characteristic equations, and explain their theoretical significance. Apply concepts of orthogonality, inner product spaces, and quadratic forms to solve mathematical problems. Use linear algebraic methods to model and solve problems arising in applied fields such as computer science, engineering, and data analysis.

Home Work: 15%

Quizzes & Class Presentation: 25%

Mid-Term: 20%

Final Exam: 40%

COURSE OVERVIEW	
Week	Topics
1	Introduction and Preliminaries
2	Systems of Linear Equations I
3	Systems of Linear Equations II
4	Algebra of Matrices
5	Determinants I
6	Determinants II and Applications
7	Rank and Systems of Equations
8	Vector Spaces
9	Linear Dependence and Independence
10	Basis and Dimension
11	Linear Transformations
12	Inner Product Spaces
13	Orthogonal Transformations
14	Eigenvalues and Eigenvectors
15	Characteristic Equations & Diagonalization
16	Quadratic Forms & Review

Textbooks/Supplementary Reading

1. . Elementary Linear Algebra by Howard Anton
2. Linear Algebra and its Applications by Gibert Strang